

Maths Mock 3 — answer booklet

Answers and marking for Papers 1, 2 and 3

Award the marks shown. For 2-mark questions the right-hand column says when to give 1 mark for a correct method. Answers to fraction questions in Paper 1 may be given in any equivalent form unless the question asks for the simplest form.

Paper 1: arithmetic (40 marks)

Q	Answer	Marks	Notes
1	790	1	
2	146	1	
3	300	1	
4	142.101	1	
5	798	1	
6	12	1	
7	5,992	1	
8	900	1	
9	71	1	
10	6	1	
11	$\frac{5}{21}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
12	90	1	
13	1,996	1	
14	14,927	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long multiplication with no more than one arithmetic error.
15	250	1	
16	16	1	
17	50.49	1	
18	$\frac{1}{10}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
19	57	1	

20	938,024	1	
21	933	1	
22	$1\frac{5}{6}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
23	50.5	1	
24	$4\frac{3}{4}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
25	85	1	
26	$\frac{3}{25}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
27	693	1	
28	157	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long division with no more than one arithmetic error.
29	$5\frac{1}{12}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
30	221,619	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long multiplication with no more than one arithmetic error.
31	45	1	
32	595	1	
33	12	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
34	274	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long division with no more than one arithmetic error.
35	136	1	
36	$4\frac{31}{42}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.

Paper 2: reasoning (35 marks)

Q	Answer	Marks	Notes
1	D	1	
2	1,350 and 1,575	1	
3	£0.46	1	
4a	(5, 2)	1	
4b	(1, 7)	1	
5a	8	1	
5b	2,788,431	1	
6	36 and 135	1	
7a	€18	1	
7b	£10	1	
8	trapezium	1	
9a	9	1	
9b	7	1	
10	19:45	1	
11	86 km	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $233 - 80 = 67$
12	Explanation that half a litre is 500 millilitres, so 400 millilitres is less than half a litre.	1	
13	$(20 \times 6) + (9 \times 6)$ and $(6 \times 30) - (6 \times 1)$	2	Award 2 marks for both correct calculations ticked and no others. Award 1 mark for one correct calculation ticked and no others, or both correct and one wrong.
14	41.3 km	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error.
15	72%	1	
16	0.5 kg	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $1 \frac{1}{5} = 1.2$, then $1.7 - 1.2$
17	£1.25	1	

18	15/16, 19/21, 16/17	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
19	£37	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $1,248 \div 16 = 78$, then $78 - 23 = 18$
20	3 and 29; 13 and 19	2	Award 2 marks for all 2 pairs and no incorrect pairs (in any order). Award 1 mark for 1 correct pairs and no incorrect pairs.
21	80 g	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. £2 is $40 \times 5\text{p}$ coins: $40 \times 3.25 = 130\text{ g}$; £2 is $10 \times 20\text{p}$ coins: $10 \times 5 = 50\text{ g}$; then subtract
22	60 cm ³	1	
23	10 km	1	

Paper 3: reasoning (35 marks)

Q	Answer	Marks	Notes
1a	1,199,000, 1,190,100	1	
1b	536,111, 536,396	1	
2	92,900	1	
3	£1.21	1	
4	(9, 1)	1	
5	3,525 kg	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $7 \times 75 = 525$, then $3,000 + 525$
6	$19/4 = 4 \frac{3}{4}$; $27/4 = 6 \frac{3}{4}$; $17/4 = 4 \frac{1}{4}$; $23/4 = 5 \frac{3}{4}$	2	Award 2 marks for all four lines correct. Award 1 mark for two or three lines correct.
7	£5.49	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error.
8a	2	1	
8b	-7	1	
9	The giraffe is more than three times as tall as the hippo. The rhino is 20 cm taller than the hippo. The elephant is more than twice as heavy as the rhino.	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
10	19:10	1	
11	9/100, 0.78, 0.782, 0.98	1	
12a	$y = 7$	1	
12b	$y = 21$	1	
13a	2,400	1	
13b	108	1	
14	20	1	
15	39 days	3	Award 3 marks for the correct answer. Award 2 marks for an appropriate complete method with no more than one arithmetic error, e.g. $52 \times 48 = 2,496$, then $2,496 \div 64$. Award 1 mark for $52 \times 48 = 2,496$ seen, or for dividing a total by 64.

16a	8,120 m	1	
16b	0.333 m	1	
17	6 cm	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $13.5 \div 9 = 1.5$, then 1.5×4
18	$\frac{2}{5}$	1	Answer must be in its simplest form.
19	70, 77	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
20	59°	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $180 - 126 = 54$, then $180 - 67 = 54$
21	72 cm^2	1	